

Andrei Baroian

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ABOUT ME

Research Assistant at [SPY Lab](#) (ETH Zurich), working with Jie Zhang, Avital Shafran, and Florian Tramèr on prompt injections through adversarially optimized noise in images. Finished my MSc thesis at SPY Lab with a prompt injection defense paper, submitted to S&P27 (soon on arXiv). Also part of the Robotics Safety Division at [ETH Robotics Club](#), working on adversarial attacks against VLA models in humanoid robots. Past research includes LLM pre-training (architectural variants) and post-training (speeding up GRPO), with smaller projects in quantization and mechanistic interpretability.

PROJECTS & RESEARCH

Prompt Injection via Adversarially Optimized Image Noise Jul 2026 – Present
[SPY Lab](#), ETH Zurich — With Jie Zhang, Avital Shafran, and Florian Tramèr. Prompt injections through adversarially optimized noise in images.

MSc Thesis: (Image) Prompt Injection Feb 2026 – Jul 2026
[SPY Lab](#), ETH Zurich — Supervised by Jie Zhang and Florian Tramèr. Worked on a prompt injection defense. Submitted to [S&P27](#), soon on arXiv.

Adversarial Attacks on VLA Models in Humanoid Robots Jan 2026 – Present
[ETH Robotics Club](#) — Robotics Safety Division. Creating adversarial attacks against Vision-Language-Action (VLA) models in humanoid robots. Investigating how visual perturbations can override task instructions and induce harmful behaviors.

Prompt Replay: Speeding Up GRPO [arXiv:2603.21177](#)
LLM RLVR post-training. An overhead-free online data selection method for GRPO that reuses and prioritizes prompts (not trajectories) to preserve on-policy optimization. Buffers medium-difficulty prompts near a 50% pass rate to maximize learning signal, Reducing zero-variance prompts and accelerating early training gains. Tested on 3B and 8B models.

Crown, Frame, Reverse: Layer-Wise Scaling Variants for LLM Pre-Training [arXiv:2509.06518](#)
Explored architectural variants that redistribute capacity across transformer layers during pre-training. Introduced three layer-wise scaling patterns using linear interpolation of FFN widths and attention head counts. Pre-trained 180M parameter models on 5B tokens; all variants converged to better performance than an equal-cost isotropic baseline.

EDUCATION

Feb 2026 – Jul 2026 **Exchange Semester — MSc Thesis at [SPY Lab](#)**, ETH Zurich, Switzerland
Supervised by Jie Zhang and Florian Tramèr.

Aug 2024 – Jul 2026 **MSc Computer Science: Artificial Intelligence**, Leiden University, The Netherlands
(Graduated, GPA: 8.5/10)
Notable grades: Seminar in Deep Reinforcement Learning (10), Deep Learning (9.0), Seminar in Deep Learning (9.0), Natural Language Processing (9.0).

Sep 2021 – Jul 2024 **BSc Entrepreneurship & Business Innovation**, Tilburg University, The Netherlands

WORK EXPERIENCE

ETH Zurich - Research Assistant, [SPY Lab](#)

Jul 2026 – Present

- Prompt injections through adversarially optimized noise in images.
- Working with Jie Zhang, Avital Shafran, and Florian Tramèr.

Leiden University - Teaching Assistant, Automated Machine Learning

Sep 2025 – Jan 2026

- Graded assignments and provided feedback.
- Guided students in selecting, understanding, and presenting research papers.

Akida, The Hague - Data & AI Engineer

Mar 2025 – Jul 2026

- Built LLM-powered pipelines (Gemini API) that turn unstructured sources into structured data products for customers (500M documents/year); owned code, tests, and Azure deployments end-to-end.
- Developed filtering and classification logic using heuristics and GenAI to detect construction projects across public-sector sources.
- Built the extraction pipeline for summarization and structured information retrieval, producing the core data product.
- Designed annotation workflows and LLM evaluation.
- Deployed to staging and production on Azure; monitored production pipelines.